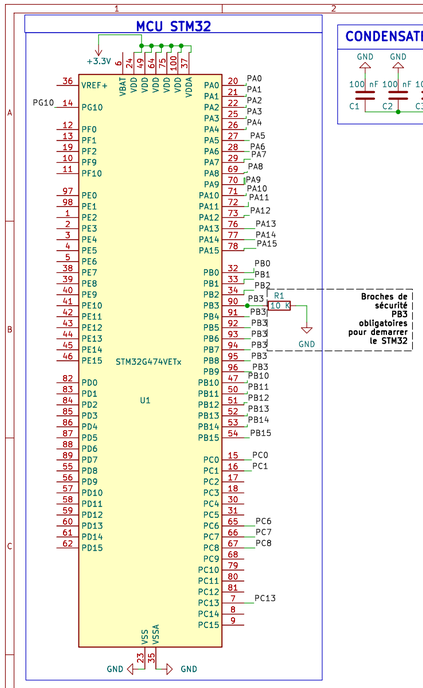
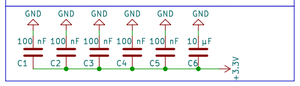
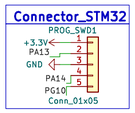
MCU STM32



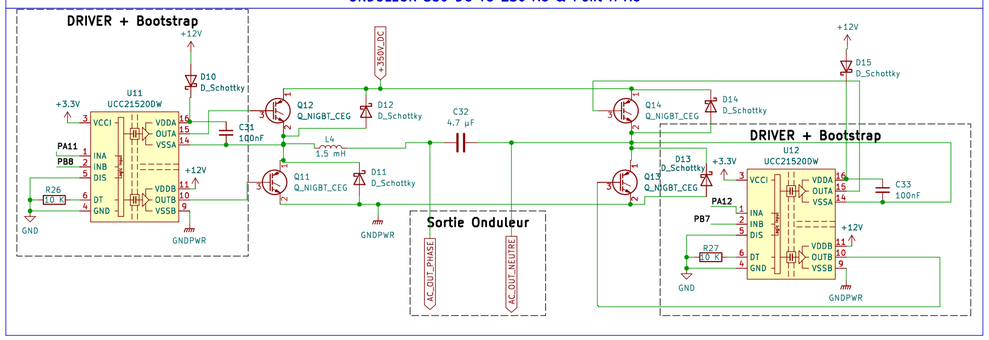
CONDENSATEUR DECOUPLAGE ALIM STM32



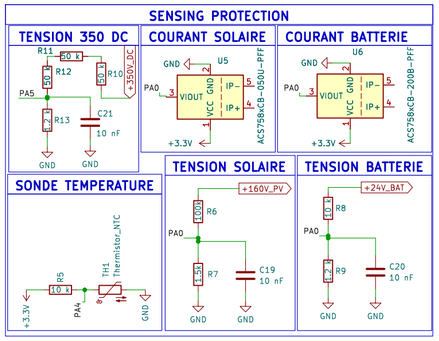
Connector_STM32



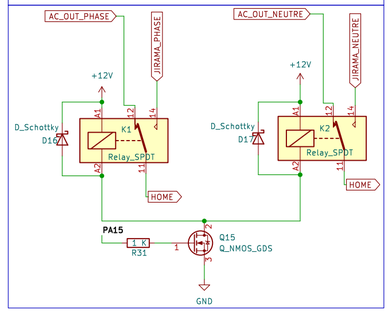
ONDULEUR 350 DC TO 230 AC & Pont H AC



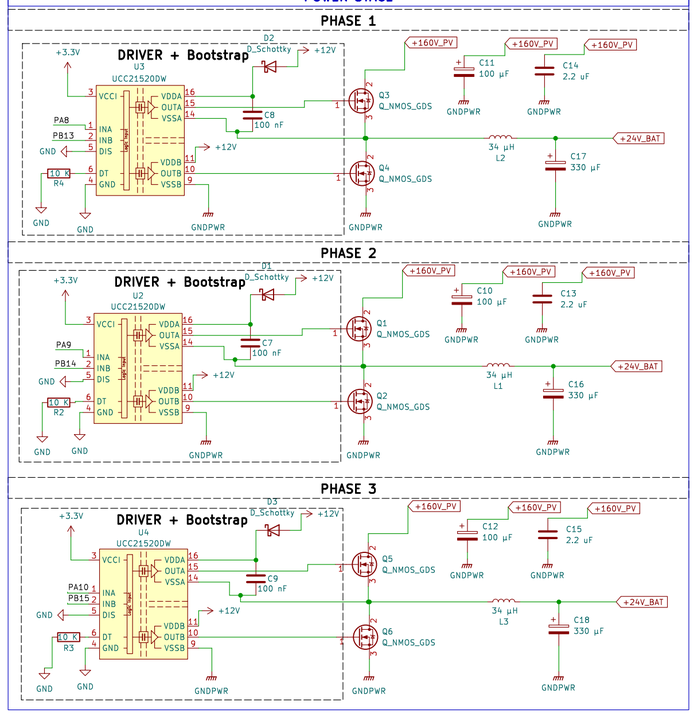
SENSING PROTECTION



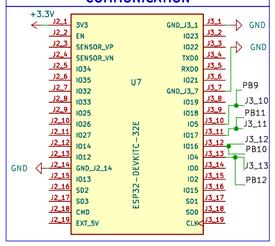
SWICTH JIRAMA & ONDULEUR



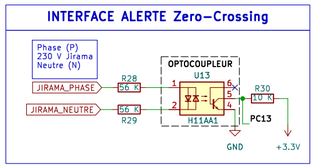
POWER STAGE



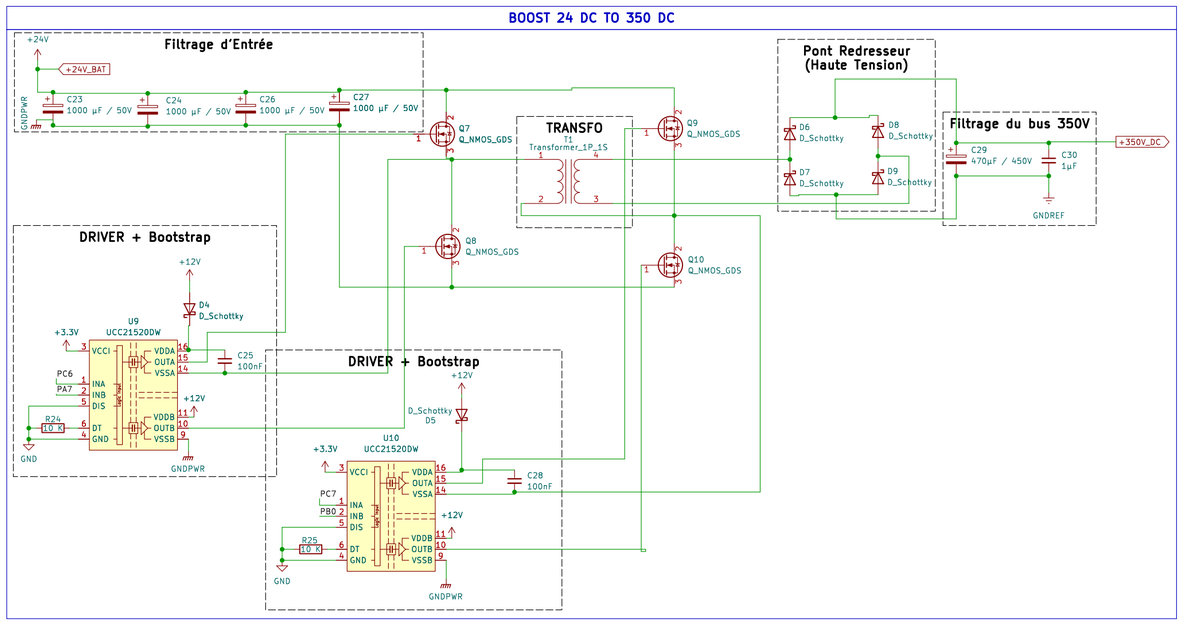
MCU ESP32 COMMUNICATION



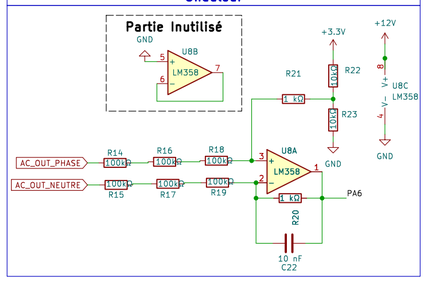
INTERFACE ALERTE Zero-Crossing



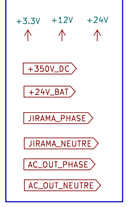
BOOST 24 DC TO 350 DC



Sensing Protection Amplificateur Opérationnel Tension Onduleur



Alimentation




```
==========================================
LEXIQUE & COMPOSANTS CLÉS DU PROJET

--- 1. ABRÉVIATIONS ESSENTIELLES ---
MPPT : Maximum Power Point Tracking (Algo d'optimisation solaire)
PWM  : Pulse Width Modulation (Signal de commande du hacheur)
ADC  : Analog to Digital Converter (Lecture analogique des sondes)
HS/LS: High-Side (Haut) / Low-Side (Bas) désignant les MOSFETs
ESR  : Equivalent Series Resistance (Résistance interne d'un condensateur)
NTC  : Negative Temperature Coefficient (Sonde thermique de sécurité)
SWD  : Serial Wire Debug (Port de programmation du microcontrôleur)

--- 2. RÔLE DES COMPOSANTS PHYSIQUES ---
STM32G474 (MCU) : "Le Cerveau". Calcule le MPPT et génère les signaux PWM.
UCC21520 (Driver): "Le Muscle". Isole le MCU et amplifie le PWM en 12V.
MOSFET (NMOS)   : "L'Interrupteur". Hache la puissance à 50 000 fois/sec.
ACS758 (Hall)   : "Les Yeux". Mesure les 180A sans contact (magnétique).
Inductance (L)  : "Le Volant d'inertie". Lisse le courant dans chaque phase.
C_ELEC (Low-ESR) : "Le Gros Réservoir". Lisse la tension pour la batterie.
C_CERA (X7R)    : "L'Éponge Rapide". Absorbe les parasites haute fréquence.
```

Sheet: /		
File: Onduleur_Hybride_230V_V2.kicad_sch		
Title:		
Size: A2	Date:	**Rev:**
KiCad E.D.A. kicad 7.0.11		Id: 1/1